

Design and Implementation of a Virtual Reality-based Electronic Voting System

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ABSTRACT

2D web/mobile voting collects ballots efficiently but provides little sense of shared procedure in formal meetings. We present a VR prototype for shareholder-style voting in a shared virtual conference room. Avatars review agenda boards and cast a single-choice ballot via in-world buttons, while an operator console configures minimum participation and approval rules and queries the final tally. A Delightex client and a lightweight server record one vote per authenticated account. Informal demos suggest clearer procedural staging than a web ballot, while scalability and election-grade security remain future work.

Index Terms: Virtual Reality, Electronic Voting, Metaverse, Shareholder Meeting, Avatar-based Interaction

1 INTRODUCTION

Many organizations now rely on electronic ballots when members join a decision process from separate locations. In typical deployments, the procedure is reduced to web forms or simple mobile screens, which are convenient for collecting choices but provide little sense of co-location or staged progression through a formal agenda.

This limitation is particularly evident in shareholder meetings: participants may connect from different cities or countries, yet the event is still expected to resemble an official assembly. Beyond usability, the perceived formality of the procedure can matter because it may influence how deliberate the act of voting feels. Prior work on educational metaverse platforms and augmented-reality tour systems suggests that three-dimensional, place-oriented environments can increase engagement and situational awareness within an ongoing activity [2, 4, 5], motivating a VR setting as a testbed for exploring such effects in voting.

To examine this idea, we implemented a prototype that runs an entire shareholder-style session inside a VR conference room. Participants join as avatars, review agenda materials in the meeting area, and move to a dedicated voting corner to cast ballots (Fig. 1). An in-world administrative console allows the operator to set quorum and approval thresholds and to inspect the final tally without leaving the immersive space (Fig. 2). Using this prototype as a case study, we discuss how spatial staging and interaction can support electronic voting and what limitations remain in the current implementation.

2 SYSTEM OVERVIEW AND METHODS

2.1 Core Concept

The prototype is designed to keep the entire shareholder meeting workflow inside a single VR conference room. Rather than fol-

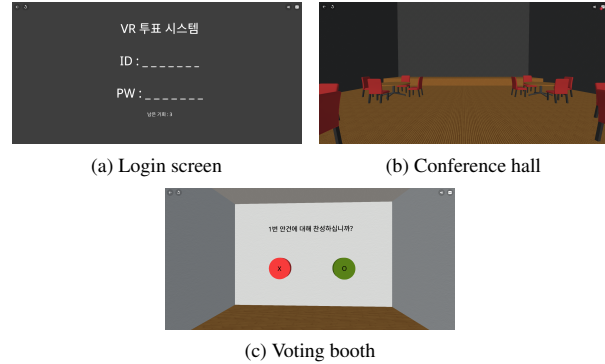


Figure 1: User interaction workflow.

lowing a series of separate web pages, participants authenticate, attend the meeting, and vote by moving between regions in the same 3D space. The transition from discussion to decision-making is therefore expressed as a change of location, not merely as a screen change.

From a software point of view, the prototype is built out of three cooperating pieces. On the client side, a Delightex-based VR application draws the login wall, meeting hall, voting corner, and the operator's console, and sends user events to the backend. A separate server process receives these events, checks login attempts and voting requests, and later computes the final counts when the session ends. Long-lived information such as account identifiers, vote history, and configuration parameters (quorum size and approval rules) is stored in a lightweight database. Communication between client and server is handled by a simple REST-like HTTP API.

2.2 User Workflow

The user-side experience is organized around movement and interaction within the virtual room.

2.2.1 Login and Entry

At the first stage, users encounter a VR login panel where they enter a login ID and password (Fig. 1(a)). The client sends the credentials to the server, which validates them and returns the authentication result. Upon success, the user is automatically placed into the main conference hall. To reduce trivial brute-force attempts, access is locked for 30 minutes after three failed login attempts.

2.2.2 Virtual Conference and Voting

After login, users find themselves in a hall with a central stage, seating, and large display surfaces (Fig. 1(b)). Agenda summaries and related materials are shown on a front screen or along the walls, supporting a presentation-like flow before voting.

Voting takes place in a dedicated area within the same room (Fig. 1(c)). The ballot is implemented as a single-choice interface with separate buttons for each option. Selecting an option stores the

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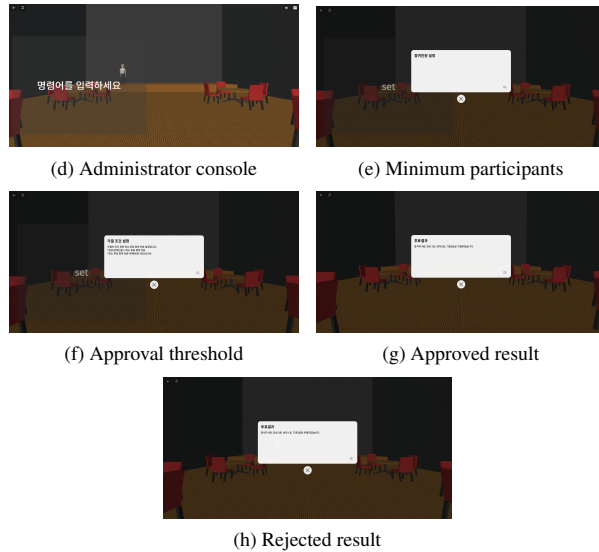


Figure 2: Administrator interface.

choice on the client and sends the selected option to the server; the server checks whether the user has already voted, validates the request, and records the first valid submission in the database. Results are presented through the administrator's result query after voting.

2.3 Administrator Console and Results

Administrators operate the voting process through an in-world console panel rather than an external dashboard.

2.3.1 Policy Configuration

The console allows the administrator to configure two policy parameters before voting begins: (1) the minimum number of participants required for the vote to be valid, and (2) the approval condition as the required number of "yes" votes for an agenda to pass (Fig. 2(d)–(f)). The entered values are sent to the server and stored as voting policy data, which are later used when determining whether the agenda is approved or rejected.

2.3.2 Result Query and Summary

To retrieve the outcome, the administrator enters a result-query command (e.g., `finvote`) in the in-world console. The request is sent to the server, which consults the vote-record table and the configuration table and returns the final summary. The result panel visualizes the total number of participants and the counts of yes/no votes, and reports whether the agenda is approved or rejected under the configured minimum participation and approval condition (Fig. 2(g) and (h)).

3 INITIAL OBSERVATIONS

Several short sessions with colleagues in the laboratory were conducted to observe how the system behaves in practice. The groups were small, and the procedure resembled an informal demo rather than a controlled user study. The remarks reported here should therefore not be interpreted as statistically reliable. Even with that caveat, certain patterns appeared repeatedly in participants' comments.

One recurring remark concerned how the layout of the room shaped the perceived flow of the meeting. Triggering the voting transition (e.g., entering the `vote` command) and being transported to the voting corner was described as a clear "step change" that

separated discussion from decision-making. Compared with simply opening a web survey in a new browser tab, placing the voting interface in a different part of the same room made the act of voting feel more deliberate and more official.

People who tried the administrator role mostly talked about the console placed inside the room. The general comment was that the panels were easy to read and that the controls for quorum settings and result querying were co-located, so there was hardly any need to switch to a separate browser window during a session.

Security functionality is also minimal in this version: beyond the login screen and a simple check that each account votes once, there is no mechanism yet for anonymity, public verification of the count or protection against coercion [1]. In practice the prototype therefore plays the role of a design sketch for interaction ideas rather than a system that could already be relied on in an actual shareholder election.

4 CONCLUSION

4.1 Summary of Contributions

This study describes a VR system that carries out a shareholder-style vote inside one continuous virtual conference room. In the prototype the procedure has three main stages: login, participation in the meeting, and the final ballot. These stages are not split across separate applications; instead, each one corresponds to a different corner of the same three-dimensional room. The operator's console is embedded in that room as well, so quorum rules can be adjusted and the final tally examined while staying inside the VR scene. Taken together, these elements show how the layout of the room and the movement of avatars can make the structure of a formal decision procedure visible to participants.

4.2 Future Work

Several extensions are planned. On the technical side, migration to a more scalable engine such as Unity is expected, so that larger groups of users and richer forms of interaction can be supported in real time. A second line of work will incorporate cryptographic techniques and blockchain-based storage to provide ballot secrecy, verifiable tallies, and stronger protection against tampering [3]. Finally, controlled comparative studies are being designed to contrast this VR prototype with conventional web-based voting interfaces, measuring perceived presence, trust, and willingness to participate in order to quantify the trade-offs of immersive voting environments.

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